

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1.(10 points) Find the derivatives of the following functions.

(a) $\sqrt[3]{x} + 2x^3$

$$\begin{aligned}\frac{d}{dx}[\sqrt[3]{x} + 2x^3] &= \frac{1}{3}x^{\frac{1}{3}-1} + 2 \cdot (3x^2) \\ &= \frac{1}{3}x^{-\frac{2}{3}} + 6x^2\end{aligned}$$

(b) $\sqrt{x}e^x$

$$\begin{aligned}\frac{d}{dx}[\sqrt{x}e^x] &= \sqrt{x} \frac{d}{dx}[e^x] + e^x \frac{d}{dx}[\sqrt{x}] \\ &= \sqrt{x}e^x + e^x \left(\frac{1}{2}x^{\frac{1}{2}-1}\right) \\ &= \sqrt{x}e^x + \frac{1}{2}x^{-\frac{1}{2}}e^x \\ &= e^x\left(\sqrt{x} + \frac{1}{2\sqrt{x}}\right)\end{aligned}$$

(c) $\frac{x}{x+2e^x}$

$$\begin{aligned}\frac{d}{dx}\left[\frac{x}{x+2e^x}\right] &= \frac{(x+2e^x)\frac{d}{dx}[x] - x \cdot \frac{d}{dx}[x+2e^x]}{(x+2e^x)^2} \\ &= \frac{x+2e^x - x(1+2e^x)}{(x+2e^x)^2} \\ &= \frac{2e^x - 2xe^x}{(x+2e^x)^2} \\ &= \frac{2e^x(1-x)}{(x+e^x)^2}\end{aligned}$$